Mohammed-260490876-2026-Mini1

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1 Mini Project – Chocolate Chilli

2 Level 1

2.1 Sajid Mohammed

2.2 260490876

2.3 14/10/26

2.4 Version 1

2.5 Summary of the Question

The goal of this project is to develop a procedural Java application which simulates a game of Chocolate Chilli between a human and a computer opponent. Players take turns removing 1, 2 or 3 chocolates from a pile. The player who eats the last chocolate wins; the player who has no chocolates at the beginning of his turn loses. As the project progresses the computer will learn from previous games and improve its decision making to increase its chances of winning. The program will introduce the game, explain the rules of the game, ask for the player's name and the initial number of chocolates, and show the above information for level 1. The program will use basic techniques and input/output commands but will not run the full game.

2.6 Justification that the program passes this level

This is based on tick boxes at start of the question description for this mini-project. - I can EXPLAIN HOW MY PROGRAM WORKS - I wrote this program myself - I use screen output to display information - I use keyboard input to collect user information - I split my code into methods that perform well-defined tasks - I use a sequence of method calls - I include my name and student number in the program comments - I document and test each method separately - I include a complete tested version of my program at the end

2.7 The literate program development

2.7.1 – displayWelcomeMessage –

What it does

Presents the user with a welcome message

Implementation (how it works) usees system out to print a welcome statement

[2]:
```
//method to display welcome message

public static void displayWelcomeMessage() {
        System.out.println("Welcome to Chocolate Chilli!");
    }
```

Testing See whether the message appears as output

[3]:
```
// TEST CODE for method 1
displayWelcomeMessage();
```

```
Welcome to Chocolate Chilli!
```

2.7.2 – explainRules –

What it does – Explains the rules of the game Chocolate Chilli to the user

Implementation (how it works) – Using multiple println statements to format the text in an appropriate readable manner

[4]:
```
//method to explain the rules of the game
    public static void explainRules() {
        System.out.println("");
        System.out.println("The rules of Chocolate Chilli are as follows:");
        System.out.println("Players take turns eating 1, 2 or 3 chocolates.");
        System.out.println("Whoever takes the last chocolate wins.");
        System.out.println("The losing player must eat the chilli!");
        System.out.println("");
    }
```

Testing – See whether the message appears as output

[7]:
```
// TEST CODE for method 2
explainRules();
```

```
The rules of Chocolate Chilli are as follows:
Players take turns eating 1, 2 or 3 chocolates.
Whoever takes the last chocolate wins.
The losing player must eat the chilli!
```

2.7.3 – setupGame –

What it does – Takes an input of the player's name and how many chocolates they would like to start the game with, and displays an output confirming the users choices

Implementation (how it works) – Displays a prompt to collect the input from user, and displays relevant output

[10]:
```
// Method 3 code
//method to set up the game
    public static void setupGame(Scanner input){
        String player1;
        System.out.print("Enter name for Player 1: ");
        player1 = input.nextLine();

        System.out.print("How many chocolates would you like to start with? ");
        String chocolates1 = input.nextLine();
        int chocolates = Integer.parseInt(chocolates1);

        System.out.println("");
        System.out.println("Thanks " + player1) ;
        System.out.println("You have " + chocolates + " chocolates to start␣
 ↪with.");
        System.out.println("I will go first!");
    }
```

Testing – Send in the Scanner method as input, test whether inputs are collected, and outputs are correctly presented

[11]:
```
// TEST CODE for method 3
Scanner input = new Scanner(System.in);
setupGame(input);
```

```
Enter name for Player 1:

 rehan

How many chocolates would you like to start with?

 11


Thanks rehan
You have 11 chocolates to start with.
I will go first!
```

2.7.4 – main() –

What it does – Groups all the modules together and runs them sequentially

Implementation (how it works) – Takes all the modules within one module and runs them in the correct order

[12]:
```
// Method 4 code
public static void main(String[] args) {
```

```
        Scanner input = new Scanner(System.in);

        displayWelcomeMessage();
        explainRules();
        setupGame(input);


    }
```

Testing Call `main()` with an empty String array and check that the program displays the welcome message and rules, asks for the player's name and starting number of chocolates, and confirms the entered information.

[18]:
```
// TEST CODE for method
main(new String[0]);
```

```
Welcome to Chocolate Chilli!

The rules of Chocolate Chilli are as follows:
Players take turns eating 1, 2 or 3 chocolates.
Whoever takes the last chocolate wins.
The losing player must eat the chilli!

Enter name for Player 1:

 saj

How many chocolates would you like to start with?

 11


Thanks saj
You have 11 chocolates to start with.
I will go first!
```

2.8 The complete program

To run the complete program outside this notebook, save it in a file named `Main.java` (matching the public class name), then compile and run it.

[20]:
```
/*
 * Chocolate Chilli Mini Project - Level 1
 * Author: Sajid Mohammed
 * Student Number: 260490876
 * Date: 14/10/2026
 * Version: 1
 * Description: Introduces the game, explains the rules,
 * and collects the player's initial setup information.
 */
```

```java
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        displayWelcomeMessage();
        explainRules();
        setupGame(input);


    }

    //method to display welcome message
    public static void displayWelcomeMessage() {
        System.out.println("Welcome to Chocolate Chilli!");
    }

    //method to explain the rules of the game
    public static void explainRules() {
        System.out.println("");
        System.out.println("The rules of Chocolate Chilli are as follows:");
        System.out.println("Players take turns eating 1, 2 or 3 chocolates.");
        System.out.println("Whoever takes the last chocolate wins.");
        System.out.println("The losing player must eat the chilli!");
        System.out.println("");
    }

    //method to set up the game
    public static void setupGame(Scanner input){
        String player1;
        System.out.print("Enter name for Player 1: ");
        player1 = input.nextLine();

        System.out.print("How many chocolates would you like to start with? ");
        String chocolates1 = input.nextLine();
        int chocolates = Integer.parseInt(chocolates1);

        System.out.println("");
        System.out.println("Thanks " + player1) ;
        System.out.println("You have " + chocolates + " chocolates to start␣
 ↪with.");
        System.out.println("I will go first!");
    }
}
```

2.9 Testing the complete program

Run the following test calls for each test in the test plan to run the complete program.

2.9.1 Test Plan: Test 1

If the following input is given by the user when asked in turn:

```
Enter name for Player 1:
 Sajid
How many chocolates would you like to start with?
 10
...
```

The program should output exactly (not including the input):

```
Welcome to Chocolate Chilli!

The rules of Chocolate Chilli are as follows:
Players take turns eating 1, 2 or 3 chocolates.
Whoever takes the last chocolate wins.
The losing player must eat the chilli!

Enter name for Player 1:
 Sajid
How many chocolates would you like to start with?
 10

Thanks Sajid
You have 10 chocolates to start with.
I will go first!
```

[22]:
```
Main.main(null)
```

```
Welcome to Chocolate Chilli!

The rules of Chocolate Chilli are as follows:
Players take turns eating 1, 2 or 3 chocolates.
Whoever takes the last chocolate wins.
The losing player must eat the chilli!

Enter name for Player 1:

 Sajid

How many chocolates would you like to start with?

 10
```

```
Thanks Sajid
You have 10 chocolates to start with.
I will go first!
```

END OF LITERATE DOCUMENT